

Course Pack - Page 6
Solutions

①

1. $x + y = 12$ ①

$$\text{Max} = x \cdot y^3 \quad \text{②}$$

Sub. ① into ②

$$\begin{aligned} \text{Max} &= (12 - y)y^3 \\ &= 12y^3 - y^4 \end{aligned}$$

$$(\text{Max})' = 36y^2 - 4y^3$$

$$(\text{Max})' = 4y^2(9 - y)$$

$$\text{set } (\text{Max})' = 0$$

$$4y^2(9 - y) = 0$$

$$\Rightarrow y = 0 ; y = 9$$

↑

inadmissible
not positive.

$$\therefore y = 9$$

$$x = 3$$

(2)

$$2. \quad x + \frac{1}{x} = \min.$$

$$(\min)' = 1 - \frac{1}{x^2}$$

$$\text{Set } (\min)' = 0$$

$$1 - \frac{1}{x^2} = 0$$

$$\frac{1}{x^2} = 1$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\text{But } x > 0 \Rightarrow x = 1$$

$$\therefore x = 1$$

(3)

$$3. \quad x - y = 150 \quad (1)$$

$$\text{Min} = x \cdot y \quad (2)$$

$$\text{Min} = x(x - 150)$$

$$\text{Min} = x^2 - 150x$$

$$(\text{Min})' = 2x - 150$$

$$\text{set } (\text{Min})' = 0$$

$$2x - 150 = 0$$

$$x = 75$$

$$\therefore x = 75$$

$$y = -75$$

(4)

$$4. \quad xy = 200 \quad (1)$$

$$\text{min} = x + 2y$$

$$\text{min} = x + 2\left(\frac{200}{x}\right)$$

$$\text{min} = x + \frac{400}{x}$$

$$(\text{min})' = 1 - \frac{400}{x^2}$$

$$\text{set } (\text{min})' = 0$$

$$1 - \frac{400}{x^2} = 0$$

$$x^2 = 400$$

$$x = \pm 20$$

* $x = -20$ is
inadmissible
not positive.

$$\therefore x = 20$$

$$y = 10$$

(5)

$$5. \quad x + y = k$$

$$\text{Min} = x^2 + y^2$$

$$\text{Min} = x^2 + (k - x)^2$$

$$\text{Min} = x^2 + k^2 - 2kx + x^2$$

$$\text{Min} = 2x^2 - 2kx + k^2$$

$$(\text{Min})' = 4x - 2k$$

$$\text{set } (\text{Min})' = 0$$

$$4x - 2k = 0$$

$$x = \frac{k}{2} \Rightarrow y = \frac{k}{2}$$

$$\therefore \text{Min} = x^2 + y^2$$

$$\text{Min} = \left(\frac{k}{2}\right)^2 + \left(\frac{k}{2}\right)^2$$

$$= \frac{2k^2}{4}$$

$$\text{Min} = \frac{k^2}{2}$$

$\therefore$ Min is at least $\frac{1}{2}k^2$.